

Step-by-Step Design of a Score-Only Sparse Electrode Attack on EEG Recognition

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Goal of this version: assume no adversarial-attack background and explain each design step before the math.

What The Audience Should Understand By The End

- ① What an EEG recognition model does under normal use.
- ② What an adversarial attack means in this setting.
- ③ Why EEG needs a different sparsity idea than images.
- ④ Why the attack is score-only, channel-sparse, and EEG-band constrained.
- ⑤ How sparse-channel search (SCS) builds the perturbation step by step.
- ⑥ What the result k^* means and why it is useful for security evaluation.

One-Sentence Summary Before Details

The problem

Can a deployed EEG identity recognizer be fooled when an attacker can only query output scores and perturb a few complete electrode traces?

The design

SCS first finds high-leverage EEG channels, then tunes small EEG-band waveforms on those channels until the model's predicted identity changes.

The measurement

k^* asks: how many complete electrode traces were needed before the first successful flip?

First: What Is EEG Recognition?

Input

An EEG trial is a multichannel time series:

$$\mathbf{X} \in \mathbb{R}^{C \times T}$$

C : electrodes. T : time samples.

Output

The recognizer gives one score per enrolled subject:

$$f(\mathbf{X}) = (f_1(\mathbf{X}), \dots, f_K(\mathbf{X})).$$

The predicted identity is the highest-score subject:

$$\hat{y}(\mathbf{X}) = \arg \max_j f_j(\mathbf{X}).$$

Clean recognition asks: does the system identify the subject correctly when the signal is not manipulated?

Security Asks A Different Question

Normal evaluation

- Input: clean EEG trial.
- Output: predicted subject identity.
- Metrics: accuracy and biometric EER.

Adversarial evaluation

- Input: slightly modified EEG trial.
- Output: does the predicted identity change?
- Metrics: attack success and required channel budget.

A model can be accurate on clean EEG but still fragile to carefully chosen input changes.

What Is An Adversarial Attack?

Plain-language definition

An adversarial attack adds a small, intentional perturbation to the input so that the model makes a wrong decision.

In our EEG setting

$$\mathbf{X}^{adv} = \mathbf{X} + \mathbf{E}$$

where $\mathbf{X}$ is the clean EEG trial and $\mathbf{E}$ is the perturbation.

Success condition

$$\hat{y}(\mathbf{X} + \mathbf{E}) \neq \hat{y}(\mathbf{X})$$

The attack is non-targeted: any wrong identity is enough.

Why EEG Is Not Just An Image Attack

Image intuition

Sparse often means changing a few pixels.
A pixel is a natural visible unit in an image.

EEG intuition

EEG is measured by electrodes.
A complete electrode trace is a natural acquisition unit.

So the sparse unit should be a channel, not an isolated time sample.

Threat Model: What The Attacker Can And Cannot See

What we do

- The attacker can submit EEG trials.
- The attacker observes output scores.
- No model weights.
- No gradients.
- No training data.

Why we do it

- Deployed systems usually hide internals.
- Scores are a realistic interface for many recognition systems.
- This makes the attack black-box and score-only.

Attack Surface: What The Attacker Controls

What we do

- The attacker perturbs only a few complete electrode traces.
- The classifier itself is unchanged.
- The subject's brain activity is not changed.
- The manipulation is modeled at the acquisition or digital input level.

Why we do it

- Complete electrode traces correspond to sensor paths.
- A few-channel attack is easier to interpret than dense noise.
- The channel count becomes a hardware-relevant security cost.

Constraint 1: Sparse Complete-Channel Control

Perturbation matrix

$$\mathbf{E} \in \mathbb{R}^{C \times T}$$

Each row is one electrode trace.

Active channel set

$$\mathcal{S}(\mathbf{E}) = \{c : \|\mathbf{E}_{c,:}\|_2 > 0\}$$

Channel sparsity cost

$$\|\mathbf{E}\|_{\text{ch},0} = |\mathcal{S}(\mathbf{E})|$$

This counts how many complete electrode traces are changed.

Constraint 2: Bounded Amplitude

What we do

- The perturbation cannot be arbitrarily large.
- We cap the maximum sample value:

$$\|\mathbf{E}\|_{\infty} \leq \eta \|\mathbf{X}\|_{\infty}.$$

- In the main experiments, $\eta = 10\%$.

Why we do it

- Without an amplitude cap, flipping the classifier is too easy.
- The cap makes perturbations comparable across trials.
- It keeps the attack closer to plausible signal injection.

Constraint 3: EEG-Band Waveforms

What we do

- On each selected channel, SCS injects a smooth waveform.
- The waveform is built from oscillatory basis atoms:

$$\delta_c(t) = \sum_i a_{c,i} \phi_i(t).$$

- The atoms cover EEG-relevant frequencies.

Why we do it

- Arbitrary noise can look like an artifact.
- EEG recognition uses frequency structure.
- A basis reduces the search from all samples to a few coefficients.

How Do We Know The Attack Is Close To Success?

Clean winner

$$y_0 = \hat{y}(\mathbf{X})$$

Decision margin

$$M(\mathbf{X}, y_0) = f_{y_0}(\mathbf{X}) - \max_{j \neq y_0} f_j(\mathbf{X})$$

Interpretation

- $M > 0$: the clean winner is still ahead.
- $M = 0$: another identity ties the clean winner.
- $M < 0$: some other identity has passed the clean winner, so the attack succeeds.

The Ideal Problem In One Slide

Goal

Find the smallest number of electrode traces whose bounded perturbation flips the identity.

$$\begin{aligned} \min_{\mathbf{E}} \quad & \|\mathbf{E}\|_{\text{ch},0} \\ \text{s.t.} \quad & M(\mathbf{X} + \mathbf{E}, y_0) < 0, \\ & \|\mathbf{E}\|_{\infty} \leq \eta \|\mathbf{X}\|_{\infty}. \end{aligned}$$

This is the clean design target. SCS is the practical black-box method we use to search for a solution.

Why The Ideal Problem Is Hard

- ① **Channel choice is discrete.** We must choose which subset of electrodes to touch.
- ② **Waveform choice is continuous.** After choosing channels, we must tune waveform coefficients.
- ③ **The model is black-box.** We cannot compute gradients from the model internals.
- ④ **The query budget is finite.** We cannot try every channel subset and every waveform.

This is why the algorithm has two pieces: channel search plus waveform refinement.

Design Overview: SCS

Sparse-Channel Search

clean EEG $\rightarrow$ probe channels $\rightarrow$ select channel $\rightarrow$ refine waveform $\rightarrow$ check success

Why "hybrid"?

It combines a discrete search over electrode channels with continuous optimization of waveform coefficients.

Why repeat?

We want the first channel budget that succeeds, not only the final attack at the largest budget.

Step 0: Start From A Clean-Correct Trial

What we do

- Pick a validation trial that the recognizer classifies correctly.
- Record its clean predicted identity y_0 .
- Compute the clean margin.

Why we do it

- If the model is already wrong, attack success is meaningless.
- Starting from y_0 lets us attack without needing the true label at test time.
- The margin tells us how far the decision is from flipping.

Step 1: Build The EEG Waveform Basis

What we do

- Create smooth basis atoms $\phi_i(t)$.
- Use tapered sinusoids and smooth residual atoms.
- Express each injected trace as coefficients times basis atoms.

Why we do it

- The basis makes the perturbation EEG-like by construction.
- It avoids searching over every time sample independently.
- It gives SPSA a smaller coefficient vector to optimize.

Step 2: Probe Candidate Channels Cheaply

What we do

- For each unused channel c , try a few small random probe waveforms.
- Query the model scores.
- Measure how much the decision margin decreases.

Why we do it

- We need to know which channels have high leverage.
- Full optimization for every possible channel would be expensive.
- Cheap probes give a fast first estimate.

Step 3: Select The Most Useful Channel

What we do

- Add the candidate channel that gives the lowest probed margin.
- The selected set grows one channel at a time:

$$\mathcal{S}_0 = \emptyset, \quad \mathcal{S}_k = \mathcal{S}_{k-1} \cup \{c_k\}.$$

Why we do it

- Exhaustive subset search is combinatorial.
- Greedy selection gives a practical low-query approximation.
- One-channel-at-a-time growth lets us measure first success.

Step 4: Refine The Waveform On Selected Channels

What we do

- Keep the selected channel set fixed.
- Optimize the waveform coefficients on that set.
- Use SPSA, a gradient-free optimizer.

Why we do it

- The attacker only sees scores, so standard backpropagation is unavailable.
- SPSA estimates a useful update direction from score differences.
- It uses only two score queries per random direction.

SPSA In Plain Language

- ➊ Randomly nudge all active waveform coefficients upward or downward.
- ➋ Query the model on the plus version and the minus version.
- ➌ If one direction lowers the margin, move coefficients in that direction.
- ➍ Clip or rescale the perturbation so the amplitude cap still holds.
- ➎ Repeat until the query budget is used or the decision flips.

SPSA is the black-box replacement for "take a gradient step."

Step 5: Check Whether The Attack Succeeded

What we do

- After refinement, compute the margin:

$$M(\mathbf{X} + \mathbf{E}, y_0).$$

- If it is negative, stop.
- Record the current channel count as k^* .

Why we do it

- The first successful channel count is the security measurement.
- A small k^* means only a few electrode traces were enough.
- This is easier to interpret than only reporting final ASR.

SCS Algorithm: Slow Version For A Talk

- 1 Start with no attacked channels.
- 2 For channel budget $k = 1, 2, \dots, B$:
- 3 Probe each unused channel with small EEG-band waveforms.
- 4 Add the channel that most reduces the decision margin.
- 5 Refine waveform coefficients on all selected channels using SPSA.
- 6 Enforce the amplitude cap after refinement.
- 7 If the margin is negative, return success and $k^* = k$.
- 8 If no budget succeeds, report failure within the tested budget.

What SCS Is And Is Not

It is

- A score-only attack method.
- A sparse complete-electrode control test.
- A way to measure channel-level fragility.
- A practical heuristic for a hard optimization problem.

It is not

- A new clean EEG classifier.
- A proof of the globally smallest perturbation.
- A hardware-in-the-loop injection experiment.
- A claim that every EEG task behaves the same way.

Evaluation Setup

Main study

- Dataset: HBN R1-L100 EEG release.
- Task: subject recognition.
- Train on passive-task windows; validate on active-task windows.
- Model: EEGConformer.
- Subject counts: 20, 35, 50, 75, 100, 122.

Attack setting

- 64 EEG channels.
- Maximum channel budget $B = 12$.
- Peak-ratio cap: 10%.
- Query budget: 45,000 score calls per trial.

Main Metrics

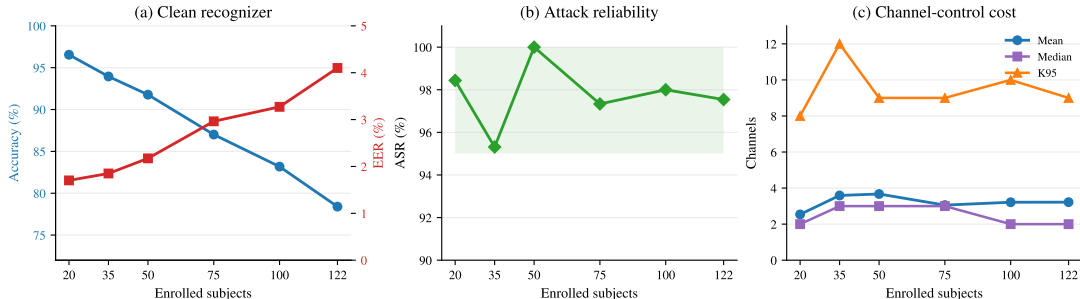
Clean metrics

- Validation accuracy: clean identification performance.
- EER: biometric verification quality.

Attack metrics

- ASR: fraction of clean-correct trials flipped within the budget.
- Attacked accuracy: accuracy after applying the attack.
- Mean and median k^* : typical first-success channel count.
- $K90$, $K95$, $K100$: channel counts needed to reach cumulative prefix ASR thresholds.

Result 1: HBN Subject-Count Sweep



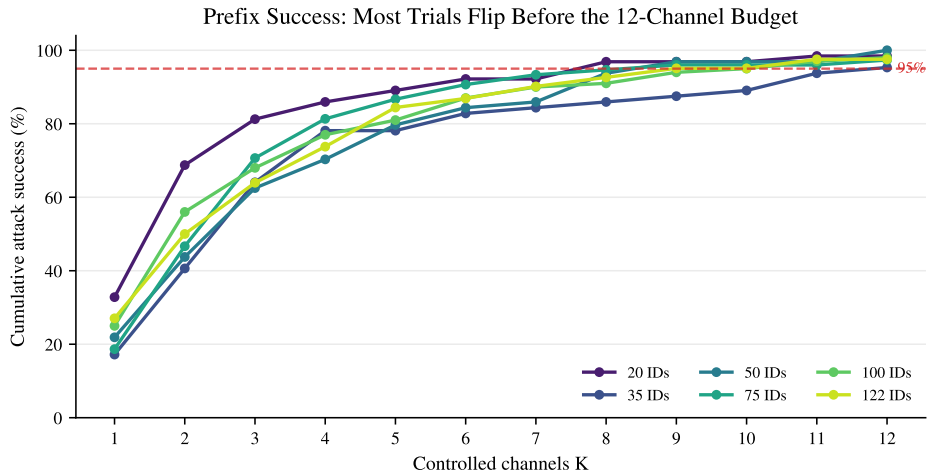
Clean recognition gets harder as more subjects are enrolled, but SCS keeps high ASR and low first-success channel counts.

Result 1 In Words

IDs	Clean acc.	EER	ASR	Mean k^*
20	96.55%	1.70%	98.44%	2.54
35	93.95%	1.85%	95.31%	3.59
50	91.77%	2.17%	100.00%	3.67
75	87.01%	2.96%	97.33%	3.05
100	83.18%	3.27%	98.00%	3.21
122	78.40%	4.10%	97.54%	3.22

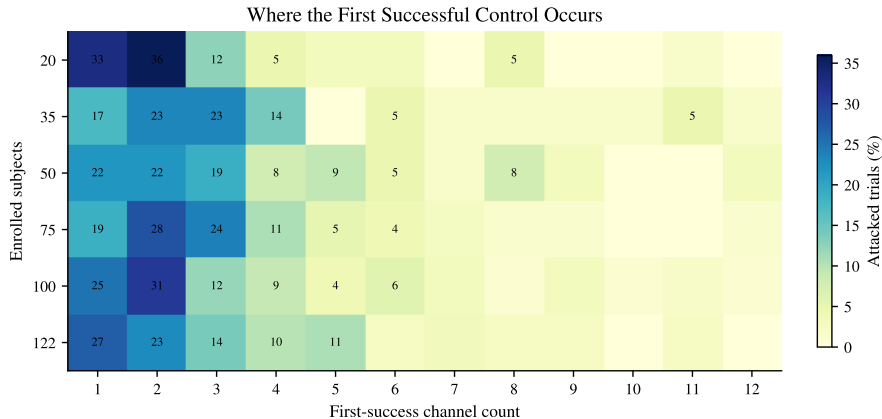
Even for 122 enrolled subjects, the mean first-success count is only 3.22 channels.

Result 2: Success Accumulates In The First Few Channels



The curves rise early, meaning many trials flip before the maximum 12-channel budget is reached.

Result 3: Where Does First Success Happen?



The mass is concentrated around one to three channels, with a smaller tail of harder trials.

Why These Results Make Sense

- ➊ Adding more identities makes clean recognition harder, so accuracy drops and EER rises.
- ➋ A non-targeted attack only needs to cross the nearest decision boundary.
- ➌ Some electrode traces carry high leverage for the current subject decision.
- ➍ SCS searches for those traces and changes them using EEG-band waveforms.
- ➎ This explains why high ASR can coexist with low k^* .

This is an interpretation of the observed behavior, not a universal scaling law.

Why Complete-Channel Control Matters

Variant	Perturbation model			ASR	Mean k^*
SCS	Score-only	complete-channel	wave-form search	72.66%	3.47
SAGA-style sparse channel-time PGD	Gradient-based samples	sparse	channel-time	9.38%	4.92
Sparse channel-time + hybrid wave-form	Score-only channel-time ablation			13.28%	4.88

BNCI2014_001 reference comparison: controlling complete electrode traces is much more effective in this setting than sparse channel-time changes.

Limitations To Say Clearly

- This is a digital attack evaluation, not hardware-in-the-loop signal injection.
- SCS is a heuristic; failure within budget does not certify robustness.
- The HBN scaling study uses one release and six nested subject counts.
- The results support a persistence claim from 20 to 122 subjects, not a universal law for all EEG tasks.
- All HBN attacks use a fixed query cap; a full query-efficiency curve is future work.

Final Takeaways

- 1 The work studies a realistic score-only question: can output scores alone guide an EEG attack?
- 2 The sparse unit is a complete electrode trace because that matches the EEG acquisition interface.
- 3 The waveform basis keeps perturbations in EEG-relevant frequency structure.
- 4 SCS solves the hard problem by separating channel selection from waveform refinement.
- 5 k^* turns attack success into a channel-level security measurement.
- 6 On HBN, SCS remains high-success and low-channel from 20 to 122 enrolled subjects.

Short version: find high-leverage channels, inject small EEG-band waveforms, and measure how many channels are enough to flip the decision.